CONNECTED HOME ENERGY FORUM · BERLIN, 26 OCTOBER 2026

# Electrification without interoperability

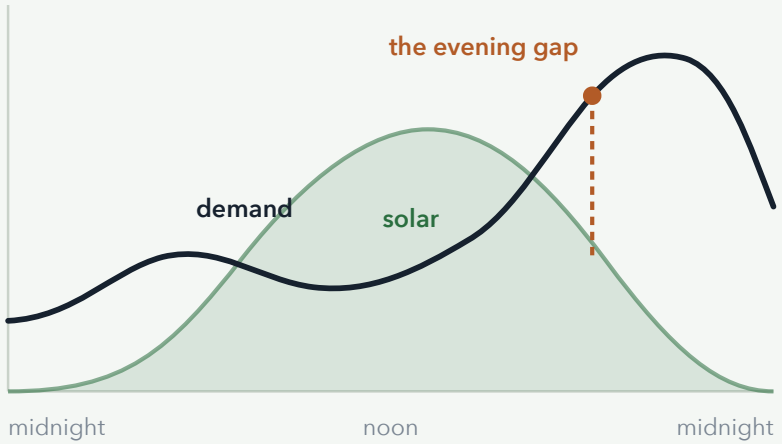
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# Electrification creates two problems. Today we are solving one.

BALANCE THE GRID

## The entire grid, hour by hour

Supply must match demand, always.

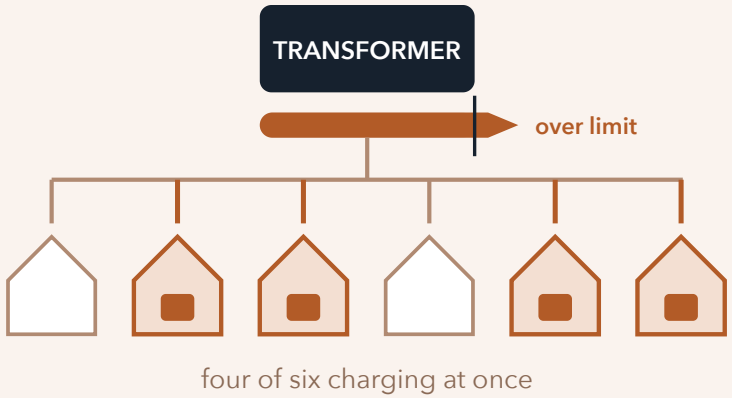



**An economics problem.** Move energy in time.

PROTECT THE TRANSFORMER

## One block, right now

The transformers and wires have a fixed capacity.




four of six charging at once

**A physics problem.** Cap power right now.

Balancing the grid is an **economics** problem. Protecting the transformer is a **physics** problem. **Both land on the customer's bill.** Peak we cannot shift is generation and transmission somebody has to build; a transformer we cannot protect is a street somebody has to dig up.

# One signal cannot do both jobs

| | A DYNAMIC PRICE | A PER-CUSTOMER POWER LIMIT |
|---|---|---|
| **What it is** | What energy costs right now | How much power this connection may draw right now |
| **Unit** | The entire grid | One service connection |
| **What it solves** | Balancing the grid | Protecting the transformer |
| **What it does** | Moves energy across time | Moves nothing. Caps power, immediately |
| **Without it** | Nobody has a reason to shift | Nothing stops everyone shifting at once |

A price is an incentive shared by everyone on the same asset. It cannot stop five houses on one transformer responding to the same number at the same moment. **A price is an incentive over a group. A limit is an obligation on one connection.**

WHAT THE DISTRIBUTION NETWORK IS ALREADY DOING

# We are already experiencing the transformer problem

**Voltage rises on a sunny afternoon**, and inverters and chargers trip off to protect themselves. The customer sees a fault, not a grid condition.

**Connections are refused, queued or derated** for heat pumps and chargers, because nobody can promise what a house will draw at the worst moment.

**Capacity is managed by curtailment and by waiting**, because there is no signal that tells a connection how much it may take right now.

> Every one of these is about protecting the transformer. **No signal in use today carries an answer to them.**

# The good news: three standards already carry a price

| | CARRIES A PRICE TODAY |
|---|---|
| **OpenADR 3** | Utility to the customer's own equipment, over the Internet |
| **Matter** | Commodity Price cluster `0x0095`, inside the home |
| **EEBUS** | Time of Use Tariff: "the absolute price SHALL be included", currency per Wh |

Tomorrow you will hear from Matter and from OpenADR on exactly this. **This half of the problem has an answer, and the answer is further along than most of this room thinks.**

NOT A PROPOSAL. A RUNNING SERVICE.

# A free, public price server, no account required

**About 1,700 programs.** Real hourly prices for two large utilities across 105 distribution circuits and substations, plus prices computed from published rate schedules.

**Hourly marginal carbon, for 11 grid regions**, on the same server, in the same shape. A greenhouse-gas number is price-shaped: a device optimizes against it exactly as it does a price.

**No authentication, no contract, no enrollment.** Open-source home automation receives it and orchestrates the house against it.

> `price.grid-coordination.energy/openadr3/3.1.0/`
> Try it in a browser during the break. Demo at the tables this evening.

# Capacity: the problem is not that nobody defined it

| | A CONNECTION-LEVEL IMPORT/EXPORT LIMIT |
|---|---|
| **OpenADR 3** | Yes, as event payloads |
| **EEBUS** | **Power Envelope**, defined at "the building's grid connection point", import and export, time-varying, with a declared fallback |
| **Matter** | `PowerThresholdStruct` is defined in the Application Cluster specification |
| **S2 / EN 50491-12-2** | No site-level construct, and the design FAQ rules out faking one by aggregating devices |
| **HCA Energy Management 1.0** | No power quantity of any kind occurs in the specification |

# The design work is done. The requirement is missing.

**This room specified the object years ago.** EEBUS defines a Power Envelope at the building's grid connection point, import and export, time-varying, with a declared fallback. That is the right object, defined in the right place, and it predates most of this conversation.

**Nothing obliges a device to receive one.** Not a standard, not a programme, not a procurement.

**What does get fielded caps the enrolled devices rather than the connection.** A limit that binds only the appliances in a scheme leaves the transformer exposed to everything else in the house.

> **The gap is not invention. It is obligation.** The objects exist in several of the standards represented in this room. Nothing requires a device to receive one, and nothing certifies that it does.

WHAT HAPPENS WHEN THE INSTRUMENT LEAVES THE INTERFACE OUT OF SCOPE

# 92.57 per cent

The **EU Code of Conduct on interoperability of energy smart appliances** declares the communication layer out of scope.

The Commission's own JRC evaluation counted **376 registered compliant products**.

| HOW THE CAPABILITY IS IMPLEMENTED | SHARE |
|---|---|
| **Manufacturer cloud digital twin** | **92.57%** |
| Built into the appliance | 6.90% |
| Dongle | 0.53% |

Published by the regulator that wrote the instrument. **Leave the interface out of scope and the market fills it with a cloud you do not control and cannot certify.** The structural answer is a standard that is local by construction, which is what makes tomorrow's Matter session worth your attention.

A CURRENT EVSE CERTIFICATION PROGRAM, SCORED AGAINST SIX TESTS

# Certification is not closing the gap either

| TEST | VERDICT | IN THE RELEASED SPECIFICATION |
|---|---|---|
| **1. An objective, not operation** | **Not met** | A Charging Schedule Function with default and **dispatch** schedules |
| **2. Open all the way to the appliance** | **Met** | Mapped onto an open protocol that terminates on the charger |
| **3. A server the customer chooses** | **Not met** | "server address" and "endpoint": zero occurrences |
| **4. Direct from the utility** | **Not met** | Aggregator mediated throughout |
| **5. A limit that holds locally** | **Not met** | "capacity limit", "import limit", "export limit": zero occurrences |
| **6. The customer who cannot respond** | **Met** | A local override, which expires every time the vehicle is unplugged |

# Yes, and also capacity

**Everything you will hear tomorrow is right, and none of it is enough on its own.** Matter makes the home coherent and does it locally: one commissioning experience across vendors, a price cluster already in the specification, and no cloud in the middle. OpenADR carries the grid's need to the customer's own equipment. Open source removes the cost of implementing both.

**The price half is converging.** Four of the standards in this room carry a price today, and one of them is serving real prices to anybody who asks, right now, with no account.

**The capacity half has objects and no obligations.** It is defined in several specifications, required by none, certified by none, and absent from the marks.

> **This is not a gap in anyone's design. It is a gap in what we ask for.**

ONE ASK

# When you write a programme, a tariff or a procurement, ask for both

**Ask for a price the device can read**, from a server the customer can point somewhere else.

**Ask for a connection limit the device must respect**, that holds when the network is down.

**Then certify that it does.** A mark that certifies dispatch and neither signal tells a buyer nothing about whether the grid can be protected.

THE ASK

**Nobody in this room has to invent anything to do this.** The objects already exist in the standards you already build on. **Put them in the next requirement you write, and the market will deliver them inside two years.**

FOR YOUR TABLES

# Four questions I would like your answers to

**What is the biggest blocker:** the objects, the obligation, or the business model?

**Who should own the connection limit?** The DNO publishes it, the customer lives under it, the device obeys it, and today no one carries it between them.

**What evidence would unlock investment?** Name the specific proof your organisation would need.

**If a limit arrived at the house tomorrow, what in your product would act on it?**

> I am facilitating a table and I will be at the demos both evenings. **Come and tell me I am wrong.**

EVERYTHING HERE IS PUBLIC AND FREE

# Where to find all of this

| | |
|---|---|
| **The price server** | `price.grid-coordination.energy/openadr3/3.1.0/` |
| **User guide, tutorials, client libraries** | `github.com/grid-coordination` |
| **The argument, in full** | `grid-coordination.energy` |
| **The six tests** | `grid-coordination.energy/presentations/` |

No account, no contract, no enrollment on any of it.